

5 k -Center

Consider the following application. Given a set of cities, with intercity distances specified, pick k cities for locating warehouses in so as to minimize the maximum distance of a city from its closest warehouse. We will study this problem, called the k -center problem, and its weighted version, under the restriction that the edge costs satisfy the triangle inequality. Without this restriction, the k -center problem cannot be approximated within factor $\alpha(n)$, for any computable function $\alpha(n)$, assuming $\mathbf{P} \neq \mathbf{NP}$ (see Exercise 5.1).

We will introduce the algorithmic technique of *parametric pruning* for solving this problem. In Chapter 17 we will use this technique in a linear programming setting.

Problem 5.1 (Metric k -center) Let $G = (V, E)$ be a complete undirected graph with edge costs satisfying the triangle inequality, and k be a positive integer. For any set $S \subseteq V$ and vertex $v \in V$, define $\text{connect}(v, S)$ to be the cost of the cheapest edge from v to a vertex in S . The problem is to find a set $S \subseteq V$, with $|S| = k$, so as to minimize $\max_v \{\text{connect}(v, S)\}$.

5.1 Parametric pruning applied to metric k -center

If we know the cost of an optimal solution, we may be able to prune away irrelevant parts of the input and thereby simplify the search for a good solution. However, as stated in Chapter 1, computing the cost of an optimal solution is precisely the difficult core of $\mathbf{NP}$ -hard $\mathbf{NP}$ -optimization problems. The technique of parametric pruning gets around this difficulty as follows. A parameter t is chosen, which can be viewed as a “guess” on the cost of an optimal solution. For each value of t , the given instance I is pruned by removing parts that will not be used in any solution of cost $> t$. Denote the pruned instance by $I(t)$. The algorithm consists of two steps. In the first step, the family of instances $I(t)$ is used for computing a lower bound on OPT , say t^* . In the second step, a solution is found in instance $I(\alpha \cdot t^*)$, for a suitable choice of α .

A restatement of the k -center problem shows how parametric pruning applies naturally to it. Sort the edges of G in nondecreasing order of cost, i.e., $\text{cost}(e_1) \leq \text{cost}(e_2) \leq \dots \leq \text{cost}(e_m)$, and let $G_i = (V, E_i)$, where $E_i =$

$\{e_1, e_2, \dots, e_i\}$. A *dominating set* in an undirected graph $H = (U, F)$ is a subset $S \subseteq U$ such that every vertex in $U - S$ is adjacent to a vertex in S . Let $\text{dom}(H)$ denote the size of a minimum cardinality dominating set in H . Computing $\text{dom}(H)$ is **NP**-hard. The k -center problem is equivalent to finding the smallest index i such that G_i has a dominating set of size at most k , i.e., G_i contains k stars spanning all vertices, where a *star* is the graph $K_{1,p}$, with $p \geq 1$. If i^* is the smallest such index, then $\text{cost}(e_{i^*})$ is the cost of an optimal k -center. We will denote this by OPT . We will work with the family of graphs $G_1, \dots, G_m$.

Define the *square of graph H* to be the graph containing an edge (u, v) whenever H has a path of length at most two between u and v , $u \neq v$. We will denote it by H^2 . The following structural result gives a method for lower bounding OPT .

Lemma 5.2 *Given a graph H , let I be an independent set in H^2 . Then, $|I| \leq \text{dom}(H)$.*

Proof: Let D be a minimum dominating set in H . Then, H contains $|D|$ stars spanning all vertices. Since each of these stars will be a clique in H^2 , H^2 contains $|D|$ cliques spanning all vertices. Clearly, I can pick at most one vertex from each clique, and the lemma follows. $\square$

The k -center algorithm is:

Algorithm 5.3 (Metric k -center)

1. Construct $G_1^2, G_2^2, \dots, G_m^2$.
2. Compute a maximal independent set, M_i , in each graph G_i^2 .
3. Find the smallest index i such that $|M_i| \leq k$, say j .
4. Return M_j .

The lower bound on which this algorithm is based is:

Lemma 5.4 *For j as defined in the algorithm, $\text{cost}(e_j) \leq \text{OPT}$.*

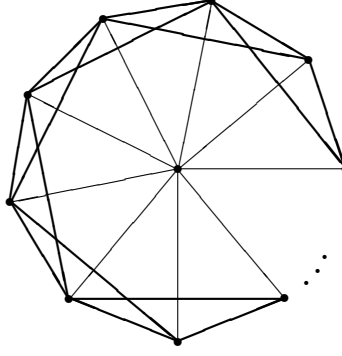
Proof: For every $i < j$ we have that $|M_i| > k$. Now, by Lemma 5.2, $\text{dom}(G_i) > k$, and so $i^* > i$. Hence, $j \leq i^*$. $\square$

Theorem 5.5 *Algorithm 5.3 achieves an approximation factor of 2 for the metric k -center problem.*

Proof: The key observation is that a maximal independent set, I , in a graph is also a dominating set (for, if some vertex v is not dominated by I , then $I \cup \{v\}$ must also be an independent set, contradicting I 's maximality). Thus,

there exist stars in G_j^2 , centered on the vertices of M_j , covering all vertices. By the triangle inequality, each edge used in constructing these stars has cost at most $2 \cdot \text{cost}(e_j)$. The theorem follows from Lemma 5.4. $\square$

Example 5.6 A tight example for the previous algorithm is given by a wheel graph on $n + 1$ vertices, where all edges incident to the center vertex have cost 1, and the rest of the edges have cost 2:



(Here, thin edges have cost 1 and thick edges have cost 2; not all edges of cost 2 are shown.)

For $k = 1$, the optimal solution is the center of the wheel, and $\text{OPT} = 1$. The algorithm will compute index $j = n$. Now, G_n^2 is a clique and, if a peripheral vertex is chosen as the maximal independent set, then the cost of the solution found is 2. $\square$

Next, we will show that 2 is essentially the best approximation factor achievable for the metric k -center problem.

Theorem 5.7 *Assuming $\mathbf{P} \neq \mathbf{NP}$, there is no polynomial time algorithm achieving a factor of $2 - \varepsilon$, $\varepsilon > 0$, for the metric k -center problem.*

Proof: We will show that such an algorithm can solve the dominating set problem in polynomial time. The idea is similar to that of Theorem 3.6 and involves giving a reduction from the dominating set problem to metric k -center. Let $G = (V, E)$, k be an instance of the dominating set problem. Construct a complete graph $G' = (V, E')$ with edge costs given by

$$\text{cost}(u, v) = \begin{cases} 1, & \text{if } (u, v) \in E, \\ 2, & \text{if } (u, v) \notin E. \end{cases}$$

Clearly, G' satisfies the triangle inequality. This reduction satisfies the conditions:

- if $\text{dom}(G) \leq k$, then G' has a k -center of cost 1, and

- if $\text{dom}(G) > k$, then the optimum cost of a k -center in G' is 2.

In the first case, when run on G' , the $(2 - \varepsilon)$ -approximation algorithm must give a solution of cost 1, since it cannot use an edge of cost 2. Hence, using this algorithm, we can distinguish between the two possibilities, thus solving the dominating set problem. $\square$

5.2 The weighted version

We will use the technique of parametric pruning to obtain a factor 3 approximation algorithm for the following generalization of the metric k -center problem.

Problem 5.8 (Metric weighted k -center) In addition to a cost function on edges, we are given a weight function on vertices, $w : V \rightarrow R^+$, and a bound $W \in R^+$. The problem is to pick $S \subseteq V$ of total weight at most W , minimizing the same objective function as before, i.e.,

$$\max_{v \in V} \{ \min_{u \in S} \{ \text{cost}(u, v) \} \}.$$

Let $\text{wdom}(G)$ denote the weight of a minimum weight dominating set in G . Then, with respect to the graphs G_i defined above, we need to find the smallest index i such that $\text{wdom}(G_i) \leq W$. If i^* is this index, then the cost of the optimal solution is $\text{OPT} = \text{cost}(e_{i^*})$.

Given a vertex weighted graph H , let I be an independent set in H^2 . For each $u \in I$, let $s(u)$ denote a lightest neighbor of u in H , where u is also considered a neighbor of itself. (Notice that the neighbor is picked in H and not in H^2 .) Let $S = \{s(u) \mid u \in I\}$. The following fact, analogous to Lemma 5.2, will be used to derive a lower bound on OPT :

Lemma 5.9 $w(S) \leq \text{wdom}(H)$.

Proof: Let D be a minimum weight dominating set of H . Then there exists a set of disjoint stars in H , centered on the vertices of D and covering all the vertices. Since each of these stars becomes a clique in H^2 , I can pick at most one vertex from each of them. Thus, each vertex in I has the center of the corresponding star available as a neighbor in H . Hence, $w(S) \leq w(D)$. $\square$

The algorithm is given below. In it, $s_i(u)$ will denote a lightest neighbor of u in G_i ; for this definition, u will also be considered a neighbor of itself.